

Co-play with Double Self: Exploring Bodily-Self Through Heautoscopy-Based XR Hide and Seek Game

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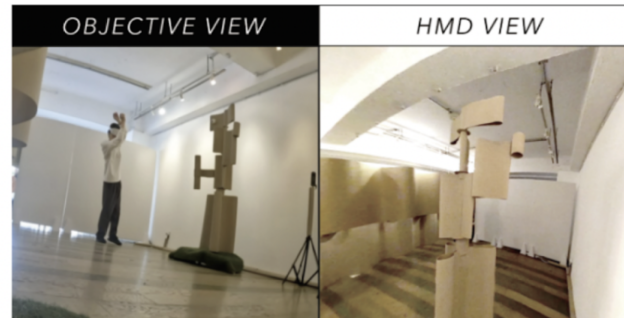
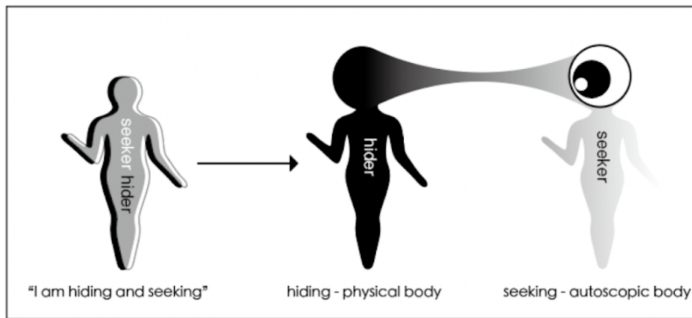


Figure 1: (left) Solo hide and seek with extracorporeal viewpoint (right) Participants locate themselves and interact with the obstacle using the extracorporeal viewpoint to fully conceal themselves within the HMD (Head-Mounted Display) vision, experiencing dual identities as both hider and seeker.

ACM Reference Format:

Kezhou Yang and Sohei Wakisaka. 2024. Co-play with Double Self: Exploring Bodily-Self Through Heautoscopy-Based XR Hide and Seek Game. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697965>

1 Introduction

Heautoscopy—the phenomenon where individuals perceive their body from a third person viewpoint like meeting one’s doppelganger—engages multiple sensory systems, including visual, proprioceptive, and vestibular. Despite its significance, existing methodologies lack effectiveness in actively inducing strong heautoscopy experiences through voluntary participant movements [Altschuler and Ramachandran 2007] [Blanke and Metzinger 2009] [Nakul et al. 2020].

This paper introduces an XR hide-and-seek game designed to induce heautoscopy experiences by integrating traditional game mechanics with an extracorporeal viewpoint. Players alternate between the roles of seeker and hider, experiencing an altered bodily self-perception through the interaction of their physical and virtual bodies. This demonstrates how VR technology can expand the boundaries of bodily awareness and reveal cognitive plasticity.

2 Approach

Unlike traditional hide-and-seek, where hiders imagine the seeker’s viewpoint, this heautoscopy-based game uses VR to visualize the seeker’s perspective while the player is hiding:

Extracorporeal Viewpoint: A 360-degree panoramic camera, wirelessly linked to the HMD, provides players with a real-time external view of their body and surroundings, disrupting the typical first-person perspective and engaging multiple sensory systems to induce heautoscopy.

Hide-and-Seek Mechanics: Players navigate physical space and use a flexible obstacle to achieve complete concealment, integrating their physical and autoscopic bodies for enhanced heautoscopy perception.

Space and Obstacle Design: The game area is designed for continuous interaction, requiring players to strategically adjust their physical positions and obstacles within distinct game areas.

Experience Flow: The game begins with a tutorial where players orient themselves by reaching an obvious box figure, helping them adapt to the bodily dislocation. In the hiding phase, players use a movable, shape-adjustable obstacle to conceal themselves from a third-person viewpoint. The challenge is to find optimal hiding spots to become completely invisible in their HMD view. The flexible obstacle design and player mobility enable dynamic strategies, fostering deeper engagement with the game’s mechanics (figure.2). To measure player performance, a scoring system was introduced where the goal is to achieve 100% invisibility in the player’s own HMD view and remain hidden for 5 seconds (figure.3). The judge confirms successful concealment using the HMD view

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SA Posters '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697965>



Figure 2: Experience flow.(top) Tutorial stage (bottom) Hiding Stage



Figure 3: Players used different strategies

on an external monitor. Players rate their performance in a post-experience questionnaire, providing insights into their strategies and effectiveness during the game.

3 Findings

The research demonstrates the formation of "double selves," where players simultaneously experience both their physical body and virtual representation. The evaluation, conducted through video recordings, verbal reports, and questionnaires, focused on three key areas related to heautoscopy [Lenggenhager et al. 2009]:

Role Alternation: Participants faced challenges in switching between the game roles of seeker and hider. This role-switching affected their timing and the strategies they employed, as they needed to continuously adjust to the sensory and physical demands of each role: The seeker role necessitated a focus on external observation, while the hider role emphasized self-concealment, prompting participants to adjust their perception of both spaces.

Bi-location: Participants engaged with obstacles to achieve complete concealment, refining their hiding strategies using the extracorporeal viewpoint. This process enhanced their sense of bi-location—creating the sensation of being simultaneously present in both their physical body and their autoscopic, virtual body, which allowed players to adapt and optimize their actions.

Bodily-Self Perception: Post-experience questionnaires revealed a heightened sense of dual presence as existing in both their physical and autoscopic bodies, which demanded multisensory integration

- "Like a **top view game**...from a third person viewpoint. It felt like in a game **controlling your own avatar**."
- "It was a sensation I had never experienced before, and it was enjoyable. I felt like **I was inside a game**."
- "At the beginning I felt I was looking at my **avatar**...When hiding I felt uncanny...I felt strongly **controlled by the camera side** (autoscopic body)."
- "Once I could see my whole body, I started to grasp my position visually. It felt similar to **controlling** a racing game from a bird's-eye view."

Figure 4: verbal reports from participants

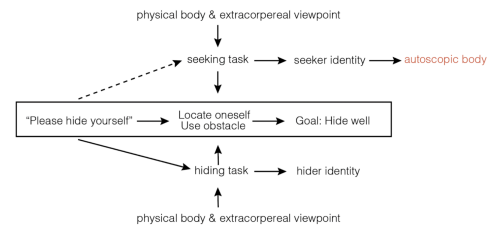


Figure 5: Dual roles, double body

and proprioceptive coordination. This led to a continuous and dynamic shift in their sense of body ownership and agency, as participants reported feeling their control and identification shift between their physical self and their perceived, externalized self (figure.4).

4 Discussion

This study shows how the dual-role dynamic in an XR hide-and-seek game enhances bodily self-perception through heautoscopy. Future work should improve evaluation methods to better capture participants' confusion and subjective experiences, especially during role-switching and self-location clarity. Applying this approach to areas like sports training—where body awareness and spatial perception are crucial, such as boxing or dance—could provide insights into motor control and proprioception. Further research should explore sensory manipulation, such as auditory feedback, to deepen understanding of multisensory integration process and cognitive plasticity, expanding applications in XR and cognitive science.

Acknowledgments

Special thanks to Saito Tatsuya and Kouta Minamizawa for their crucial contributions and support.

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